# Algorithms
# Chapter 4 Divide-and-Conquer

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# Outline

- **The substitution method**
- The recursion-tree method
- The master method
- The maximum-subarray problem
- Strassen's algorithm for matrix multiplication

# The purpose of this chapter

- When an algorithm contains a recursive call to itself, its running time can often be described by a recurrence.
- A **recurrence** is an equation or inequality that describes a function in terms of its value on small inputs.
  - For example: the worst-case running time of Merge-Sort

$$T(n)=\begin{cases}\theta(1) & \text{if } n=1,\\ 2T(n/2)+\theta(n) & \text{if } n>1.\end{cases}$$

  - The solution is $T(n)=\Theta(n\lg n)$.
- Three methods for solving recurrences 三種方法
  - the substitution method 置換法
  - the recursion-tree method 遞迴樹
  - the master method 大師方法(公式)

# Technicalities$_{1/2}$ T(n)：只定義n是整數時

- The running time $T(n)$ is only defined when $n$ is an integer, since the size of the input is always an integer for most algorithms.
  - For example: the running time of Merge-Sort is really

$$T(n)=\begin{cases}\theta(1) & \text{if } n=1,\\ T(\lceil n/2\rceil)+T(\lfloor n/2\rfloor)+\theta(n) & \text{if } n>1.\end{cases}$$

- Typically, we ignore the boundary conditions. 忽略邊界條件
- Since the running time of an algorithm on a constant-sized input is a constant, we have $T(n) = \Theta(1)$ for sufficiently small $n$.
- Thus, we can rewrite the recurrence as

當n夠小時，$T(n)=\theta(1)$

忽略 ceiling 與 floor

$$T(n) = 2T(n/2)+ \Theta(n).$$

# Technicalities$_{2/2}$ 在計算時，忽略floors，ceiling以及boundary conditions

- When we state and solve recurrences, we often omit floors, ceilings, and boundary conditions.
- We forge ahead without these details and later determine whether or not they matter. 先忽略細節，再回頭看是否重要
- They usually don't, but it is important to know when they do.
  通常不重要，但知道背後原因是重要的

# The substitution method$_{1/3}$ 置換法

- The substitution method entails two steps: 2步驟
  - Guess the form of the solution 依照經驗猜答案
  - Use mathematical induction to find the constants and show that the solution works 用數歸法證明是對的[把常數找出來]
- This method is powerful, but it obviously can be applied key only in cases when it is easy to guess the form of the answer

功能強大，但只對答案容易猜時有用

# The substitution method$_{2/3}$

- For example: determine an upper bound on the recurrence

$$T(n) = 2T(\lfloor n/2 \rfloor) + n, \text{ where } T(1)=1.$$

  - guess that the solution is $T(n)=O(n \lg n)$ 依照經驗猜答案
  - prove that there exist positive constants $c>0$ and $n_0$ such that $T(n) \le cn \lg n$ for all $n \ge n_0$ 證明 big O 的要件[找 c 與 $n_0$ 兩個常數]
  - **Basis step**:
    - when $n=1$, $T(1) \le c_1 1 \lg 1 = 0$, which is at odds with $T(1)=1$ $1 \le c_1 \cdot 0$ $\Rightarrow$ 所以 $c_1$ 不存在
    - 找 $n_0=2$ Basis: 對於 $n=2,3$ 成立
    - since the recurrence does not depend directly on $T(1)$, we can replace $T(1)$ by $T(2)=4$ and $T(3)=5$ as the base cases $T(2)=2T(1)+2=2\cdot 1+2=4$ $T(3)=2T(1)+3=2\cdot 1+3=5$
    - $T(2) \le c_1 2 \lg 2$ and $T(3) \le c_1 3 \lg 3$ for any choice of $c_1 \ge 2$
    - thus, we can choose $c_1=2$ and $n_0=2$ $4 \le c_1 \cdot 2$, $5 \le c_1 \cdot 3 \cdot 1 \ldots$ $\Rightarrow c_1 \ge 2$ 可以滿足

∵ n ≥ 2, 在做除以2取floor後，一定會遇到 2 or 3

# The substitution method$_{3/3}$

假設 $T(\lfloor \frac{n}{2} \rfloor)$ 成立
證明 T(n) 成立

→ if 變成 $T(\lfloor \frac{n}{3} \rfloor)$
basis: 2、3、4、5
6開始 → 找T(2)

- **Induction step**:
  - assume $T(\lfloor n/2 \rfloor) \le c_2 \lfloor n/2 \rfloor \lg(\lfloor n/2 \rfloor)$ for $\lfloor n/2 \rfloor$
  - then, $T(n) = 2T(\lfloor n/2 \rfloor) + n$

置換 $\le 2(c_2 \lfloor n/2 \rfloor \lg(\lfloor n/2 \rfloor)) + n$

$\le c_2 n \lg(n/2) + n$

$= c_2 n \lg n - c_2 n \lg 2 + n$

算出的結果 $= c_2 n \lg n - c_2 n + n$

目標 $\le c_2 n \lg n$

  - the last step holds as long as $c_2 \ge 1$

⇒ 觀察後得 $c_2 \ge 1$

T(2) is true = basis
T(3) is true = basis
T(4) is true = T(2) is true + induction step
T(5) is true = T(2) is true + induction step
T(6) is true = T(3) is true + induction step
T(7) is true = T(3) is true + induction step
T(8) is true = T(4) is true + induction step
⋮
T(n) is true = $T(\lfloor \frac{n}{2} \rfloor)$ is true + induction step

- There exist positive constants $c = \max\{2, 1\}$ and $n_0 = 2$ such that $T(n) \le cn \lg n$ for all $n \ge n_0$

$c_1$ $c_2$

$c = 2$, $n_0 = 2$ ⇒ 得證

# Making a good guess 如何猜出答案

- **Experience**: $T(n)=2T(\lfloor n/2\rfloor+17)+n$
  經驗
  - when $n$ is large, the difference between $T(\lfloor n/2\rfloor)$ and $T(\lfloor n/2\rfloor+17)$ is not that large: both cut $n$ nearly evenly in half.
  - we make the guess that $T(n) = O(n\lg n)$ 當n很大時，17就顯得相對不重要
- **Loose upper and lower bounds**: $T(n)=2T(\lfloor n/2\rfloor)+n$
  - prove a lower bound of $T(n) = \Omega(n)$ and an upper bound of $T(n) = O(n^2)$ 分別先猜上界和下界 慢慢將上界往下，下界往上
  - gradually lower the upper bound and raise the lower bound until we converge on the tight solution of $T(n) = \Theta(n\lg n)$

目標

$O(n^2)$ ↓

$\theta(n\log n)$

↑ $\Omega(n)$

- **Recursion trees**:
  - will be introduced later 第二節

# Subtleties 小細節

- Consider the recurrence:

$$T(n) = T(\lfloor n/2 \rfloor) + T(\lceil n/2 \rceil) + 1$$

  - **guess** that the solution is $O(n)$, i.e., $T(n) \le cn$
  - then, $T(n) \le c\lfloor n/2 \rfloor + (c\lceil n/2 \rceil + 1$ 置換

    $= cn + 1$ 算出的結果
  - $c$ does not exist $cn+1 \nleq cn \Rightarrow c$ 不存在
  - **new guess** $T(n) \le cn - b$ 再猜
  - then, $T(n) \le (c\lfloor n/2 \rfloor - b) + (c\lceil n/2 \rceil - b) + 1$

    $= cn - 2b + 1$

    $\le cn - b$ for $b \ge 1$

    證明一個更嚴格的條件來猜到常數
  - also, the constant $c$ must be chosen large enough to handle the boundary conditions

    選擇c時，要滿足basis條件

# Avoiding pitfalls 避免犯錯

- It is easy to err in the use of asymptotic notation.
- For example: $T(n) = 2T(\lfloor n/2 \rfloor) + n$
  - guess that the solution is $O(n)$, i.e., $T(n) \leq cn$
  - then, $T(n) = 2T(\lfloor n/2 \rfloor) + n$
    $\leq 2(c\lfloor n/2 \rfloor) + n$ 置換
    $\leq cn+n$ 算出的結果
    $= O(n)$ ← **wrong** c還沒找出來[事實上c不存在]
    $\leq cn$ 目標　　$cn+n \nleq cn$
  - the error is that we haven't proved the **exact form** of the inductive hypothesis, that is, that $T(n) \leq cn$

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- **The recursion-tree method** 用遞迴樹
- The master method
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用遞迴樹產生答案，用置換法驗證

# The recursion-tree method

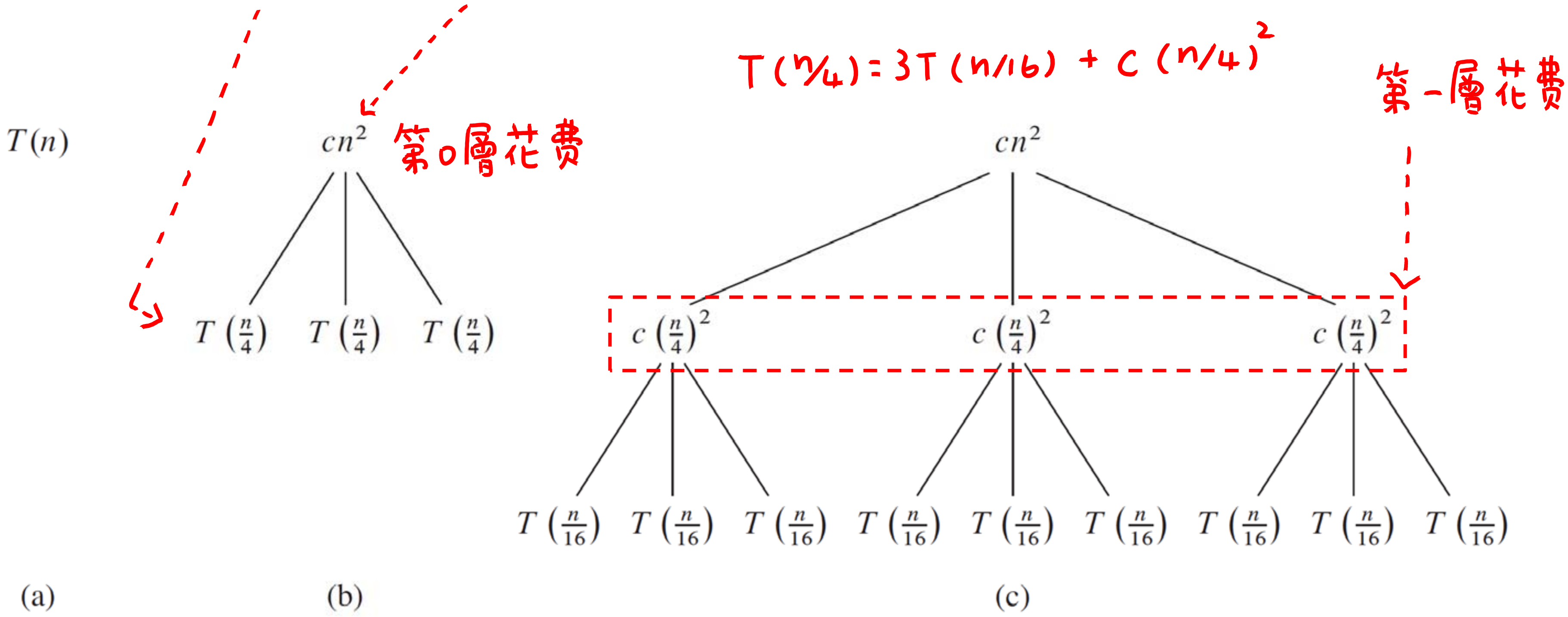
- A recursion tree is best used to generate a good guess, which is then verified by the substitution method.
- Tolerating a small amount of "sloppiness" 忽略細節, we could use recursion-tree to generate a good guess.
- One can also use a recursion tree as a direct proof of a solution to a recurrence. 也可用遞迴樹證明
- Ideas:
  - in a **recursion tree**, each node represents the cost of a single subproblem 點代表 cost
  - sum the costs within each level to obtain a set of per-level costs 將同一層花費加起來
  - sum all the per-level costs to determine the total cost 將每一層花費加起來

# An example

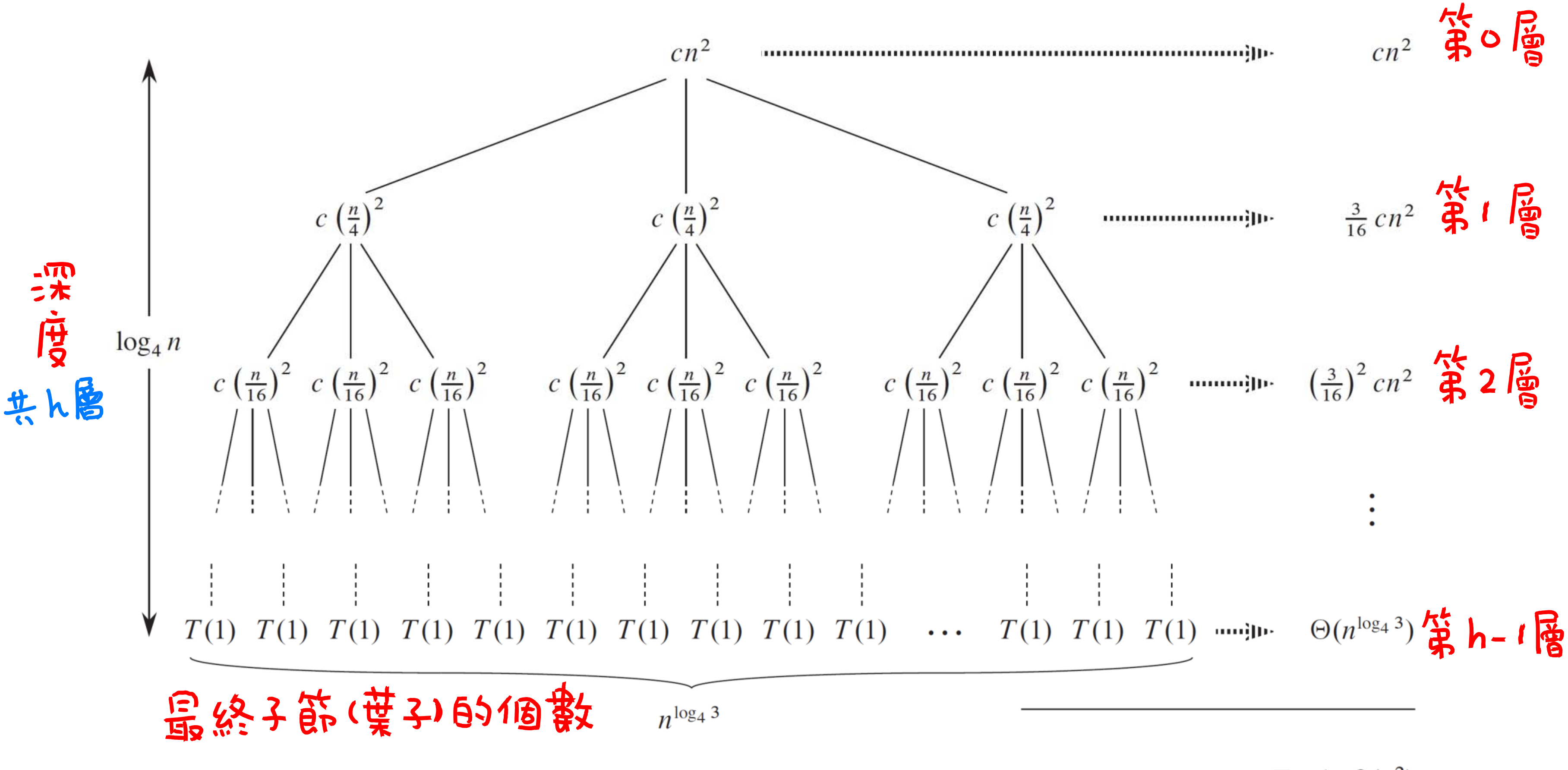
- For example: $T(n) = 3T(\lfloor n/4 \rfloor) + \Theta(n^2)$
- Tolerating the sloppiness:
  - ignore the floor in the recurrence 將floor忽略
  - assume $n$ is an exact power of 4 $n = 4^k$
- Rewrite the recurrence as $T(n) = 3T(n/4) + cn^2$

# The construction of a recursion tree$_{1/2}$

- $T(n) = 3T(n/4) + cn^2$



# The construction of a recursion tree$_{2/2}$




(d)

# Determine the cost of the tree$_{1/2}$

共h層

第i層 0 1 2 … h-1

$n \quad \frac{n}{4} \quad \frac{n}{16} \quad \cdots \quad \frac{n}{4^{h-1}} = 1$

$4^{h-1} = n$, $h = \log_4 n + 1$

- The subproblem size for a node at depth *i* is $n/4^i$.
- Thus, the tree has $\log_4 n + 1$ levels (0, 1, 2,..., $\log_4 n$).
- Each node at depth *i*, has a cost of $c(n/4^i)^2$ for $0 \le i \le \log_4 n$.
- So, the total cost over all nodes at depth *i* is $3^i * c(n/4^i)^2$ $=(3/16)^i cn^2$. 第i層的花費

  node數 每個node的花費
- The last level, at $\log_4 n$, has $3^{\log_4 n} = n^{\log_4 3}$ nodes.

  $(3^i)$ 最後一層的點數
- The cost of the entire tree:

等比級數

$$T(n) = cn^2 + \frac{3}{16}cn^2 + (\frac{3}{16})^2 cn^2 + \ldots + (\frac{3}{16})^{\log_4 n-1} cn^2 + (\frac{3}{16})^{\log_4 n} cn^2$$

$$= \sum_{i=0}^{\log_4 n} (\frac{3}{16})^i cn^2$$

$$= \boxed{\frac{(3/16)^{\log_4 n+1} - 1}{(3/16) - 1}} cn^2$$

最後一層花費 $= n^{\log_4 \frac{3}{16}} \cdot cn^2$

$= n^{\log_4 3 - \log_4 16} \cdot cn^2$

$= cn^{\log_4 3}$

$=$ 最後一層的點數 $\cdot c$

最多最多大概 $\frac{16}{13}$ 常數倍

# Determine the cost of the tree$_{2/2}$

- Take advantage of small amounts of sloppiness, we have

多算一些 → ∞

$$T(n) = \sum_{i=0}^{\log_4 n} \left(\frac{3}{16}\right)^i cn^2$$

$$< \sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 = \frac{1}{1-(3/16)} cn^2$$

$$= \frac{16}{13} cn^2 = O(n^2)$$

- Thus, we have derived a guess of $T(n)=O(n^2)$.

遞迴樹產生的答案
會忽略細節，所算出的答案非準確

# Verify the correctness of our guess 用置換法證明它正確

- Now we can use the substitution method to verify that our guess is correct.
- We want to show that $T(n) \le dn^2$ for some constant $d > 0$.
- Using the same constant $c > 0$ as before, we have

$$
\begin{aligned}
T(n) &= 3T(\lfloor n/4 \rfloor) + \theta(n^2) \\
&\le 3T(\lfloor n/4 \rfloor) + cn^2 \\
&\le 3d\lfloor n/4 \rfloor^2 + cn^2 \quad \text{置換} \\
&\le 3d(n/4)^2 + cn^2 \\
&= 3/16dn^2 + cn^2 \quad \text{算出的結果} \\
&\le dn^2, \quad \text{目標}
\end{aligned}
$$

$\theta(n^2)$
$\Rightarrow c_1 n^2 \le f(n) \le c_2 n^2$

where the last step holds for $d \ge (16/13)c$.

# Another example

分成二部份　divide and combine 時間

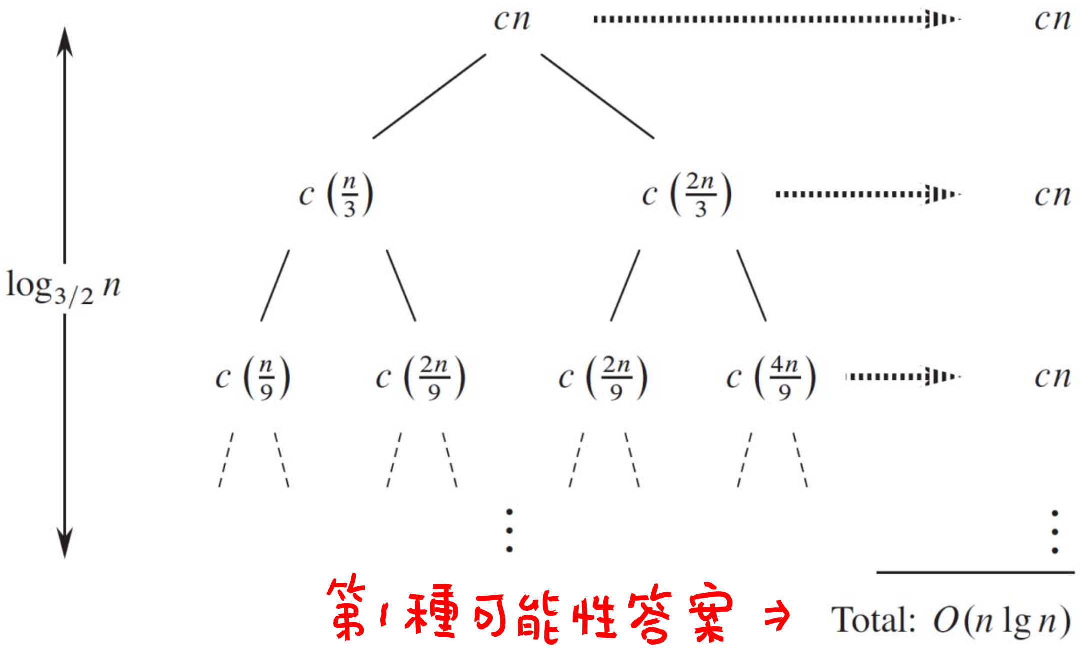
- Another example: $T(n) = T(n/3)+T(2n/3)+O(n)$.
- The recursion-tree:

共h層

第i層　$0 \quad 1 \quad 2 \quad \cdots \quad h-1$

$n \quad \frac{n}{(\frac{3}{2})^1} \quad \frac{n}{(\frac{3}{2})^2} \quad \cdots \quad \frac{n}{(\frac{3}{2})^{h-1}} = 1$

$(\frac{3}{2})^{h-1} = n, \quad h = \log_{\frac{3}{2}} n + 1$




cost: $\log_{\frac{3}{2}} n \times cn$

每一層都花 cn

第1種可能性答案 →

# Determine the cost of the tree

- In the figure, we get a value of *cn* for every level.
- The height of tree is $\log_{3/2} n$. ← 樹高 $\log_{\frac{3}{2}} n$
- Intuitively, we expect the solution to the recurrence to be at most $O(cn\log_{3/2}n) = O(n\lg n)$. ← 第1種可能性答案
- The recursion tree has fewer than $2^{\log_{3/2} n} = n^{\log_{3/2} 2}$ leaves.
- The total cost of all leaves would then be $\theta(n^{\log_{3/2} 2})$, which is $\omega(n\lg n)$. ← 第2種可能性答案
- Also, not all levels contribute a cost of exactly *cn*.

  不是每一層都花 cn 有些 leaf 尚未到最後一層就消失了(結束)
- Thus, we derived a guess of $T(n)=O(n\lg n)$.

所以,還是猜 $T(n) = O(n\log n)$

共有 $h = \log_{\frac{3}{2}} n+1$ 層 ⇒ 樹高 $= \log_{\frac{3}{2}} n$

⇒ 最後一層 leaf 個數 $< 2^{\log_{\frac{3}{2}} n} = n^{\log_{\frac{3}{2}} 2}$

⇒ 比常數倍的 $n\log n$ 大

$(n^{\varepsilon} > \log n,\ \varepsilon > 0,\ n \to \infty)$

假設 $T(n/3)$ 和 $T(2n/3)$ 成立

假設 $T(n/3) \le d(n/3)\lg(n/3)$ 和 $T(2n/3) \le d(2n/3)\lg(2n/3)$

# Verify the correctness of our guess

- We can verify the guess by the substitution method.
- We have

$$
\begin{aligned}
T(n) &= T(n/3)+T(2n/3)+O(n) \\
&\le T(n/3)+T(2n/3)+cn \\
&\le d(n/3)\lg(n/3)+d(2n/3)\lg(2n/3)+cn \quad \text{置換} \\
&= (d(n/3)\lg n-d(n/3)\lg 3) \\
&\quad +(d(2n/3)\lg n-d(2n/3)\lg(3/2))+cn \\
&= dn\lg n-d((n/3)\lg 3+(2n/3)\lg(3/2))+cn \\
&= dn\lg n-d((n/3)\lg 3+(2n/3)\lg 3-(2n/3)\lg 2)+cn \\
&= dn\lg n-dn(\lg 3-2/3)+cn \quad \text{算出的結果} \\
&\le dn\lg n \quad \text{目標} \Rightarrow \text{求 } d=?
\end{aligned}
$$

$\Rightarrow$ $d$ 的範圍

for $d \ge c/(\lg 3-(2/3))$.

recursion-tree method 只能夠猜大概
最後還是要用置換法求精確。

# Outline

- The substitution method
- The recursion-tree method
- **The master method**
- The maximum-subarray problem
- Strassen’s algorithm for matrix multiplication

# The master method$_{1/2}$ 大師方法給一個公式解

- The master method provides a "cookbook" method for solving recurrences of the form

a個子問題，每一個大小 $\frac{n}{b}$

$T(\frac{n}{b})$ ⇒ 子問題所需時間

$f(n)$ = 分割和合併所需時間

$$T(n) = aT(n/b) + f(n)$$

  - $a \geq 1$ and $b > 1$ are constants
  - $f(n)$ is an asymptotically positive function n 夠大時，f(n) 值為正

- It requires memorization of three cases, but then the solution of many recurrences can be determined quite easily.

三種情形，可解大多數的 recurrences

⇒ 有一些不能解

# The master method$_{2/2}$

- The recurrence $T(n) = aT(n/b) + f(n)$ describes the running time of an algorithm that
  - divides a problem of size $n$ into $a$ subproblems, each of size $n/b$
  - each of subproblems is solved recursively in time $T(n/b)$
  - the cost of dividing and combining the results is $f(n)$
- For example, the recurrence arising from the MERGE-SORT procedure has $a = 2$, $b = 2$, and $f(n) = \Theta(n)$. $f(n) = \underbrace{\text{divide}}_{\Theta(1)} + \underbrace{\text{merge}}_{\Theta(n)}$

  Merge-Sort：分割成2個子問題，子問題大小為$\frac{n}{2}$
- Normally, we omit the floor and ceiling functions when writing divide-and-conquer recurrences of this form.

  忽略 ceiling 和 floor

# Master theorem $n^{\log_b a}$ v.s. $f(n)$ 比大小 ⇒誰大時間複雜度就為誰

- **Master theorem:** Let $a \geq 1$ and $b > 1$ be constants, let $f(n)$ be a function, and let $T(n)$ be defined on the nonnegative integers by the recurrence

$$T(n) = aT(n/b) + f(n)$$

可以是 $\lfloor n/b \rfloor$ 或 $\lceil n/b \rceil$

where we interpret $n/b$ to mean either $\lfloor n/b \rfloor$ or $\lceil n/b \rceil$. Then, $T(n)$ can be bounded asymptotically as follows.

$n^{\log_b a}$ 夠大，相差 $n^{\varepsilon}$ ⇒ $T(n)=\theta(n^{\log_b a})$

1. If $f(n) = O(n^{\log_b a-\varepsilon})$ for some constant $\varepsilon > 0$, then $T(n) = \theta(n^{\log_b a})$.
2. If $f(n) = \theta(n^{\log_b a})$, then $T(n) = \theta(n^{\log_b a} \lg n)$. 相等，乘以 $\lg n$
3. If $f(n) = \Omega(n^{\log_b a+\varepsilon})$ for some constant $\varepsilon > 0$, and if $af(n/b) \leq cf(n)$ for some constant $c<1$ and all sufficiently large $n$, then $T(n) = \theta(f(n))$. $n^{\log_b a}$ 小，相差 $n^{\varepsilon}$ 且滿足正規情況

⇒ $T(n) = \theta(f(n))$

# Intuition behind the master method

- Intuitively, the solution to the recurrence is determined by comparing the two functions $f(n)$ and $n^{\log_b a}$.
  - Case 1: if $n^{\log_b a}$ is asymptotically larger than $f(n)$ by a factor of $n^{\varepsilon}$ for some constant $\varepsilon > 0$, then the solution is $T(n) = \theta(n^{\log_b a})$ .
  - Case 2: if $n^{\log_b a}$ is asymptotically equal to $f(n)$, then the solution is $T(n) = \theta(n^{\log_b a} \lg n)$.
  - Case 3: if $n^{\log_b a}$ is asymptotically smaller then $f(n)$ by a factor of $n^{\varepsilon}$, and the function $f(n)$ satisfies the "regularity" condition that $af(n/b) \le cf(n)$, then the solution is $T(n) = \theta(f(n))$.
- The three cases do not cover all the possibilities for $f(n)$.

三種情形，可解大多數的recurrences
⇒有一些不能解

# Using the master method$_{1/3}$

- Example 1: $T(n) = 9T(n/3) + n$

  $n \cdot n^{\varepsilon} = n^2$

  $n^{\log_b a}$ 較大, $\varepsilon = 1$

  - For this recurrence, we have $a = 9$, $b = 3$, $f(n) = n$.
  - Thus, $n^{\log_b a} = n^{\log_3 9} = \theta(n^2)$.
  - Since $f(n) = O(n^{\log_3 9 - \varepsilon})$, where $\varepsilon = 1$, we can apply case 1.
  - The solution is $T(n) = \Theta(n^2)$.

- Example 2: $T(n) = T(2n/3) + 1$
  - For this recurrence, we have $a = 1$, $b = 3/2$, $f(n) = 1$.
  - Thus, $n^{\log_b a} = n^{\log_{3/2} 1} = n^0 = 1$.
  - Since $f(n) = O(n^{\log_b a}) = \theta(1)$, we can apply case 2.
  - The solution is $T(n) = \Theta(\lg n)$.

  相等, $n^{\log_b a} \times \lg n = 1 \times \lg n = \lg n$

要大超過 $n^{\varepsilon}$

$n^{\log_b a} = n$ , $f(n) = n\,\lg n$

$n < n\lg n$ ，但是不夠大

[只有大於 $\lg n$，不到 $n^{\varepsilon}$]  $n^{\varepsilon} > \lg n$

$\lim_{n\to\infty} \frac{\log_b n}{n^a} = 0$ , for $a>0$ , $b \geq 1$

# Using the master method$_{3/3}$

- Example 4: $T(n) = 2T(n/2) + n\lg n$
  - For this recurrence, we have $a = 2$, $b = 2$, $f(n) = n\lg n$.
  - The function $f(n) = n\lg n$ is asymptotically larger than $n^{\log_b a} = n^{\log_2 2} = n$.
  - But, it is not polynomially larger since the ratio $f(n)/n^{\log_b a} = (n\lg n)/n = \lg n$ is asymptotically less than $n^{\varepsilon}$ for any positive constant $\varepsilon$.
  - Consequently, the recurrence falls into the gap between case 2 and case 3. 遞迴式落在 case 2 和 case 3 之間
- If $g(n)$ is asymptotically larger than $f(n)$ by a factor of $n^{\varepsilon}$ for some constant $\varepsilon>0$, then we said $g(n)$ is **polynomially larger** than $f(n)$.

$f(n)$ 大 $\begin{cases} \text{大於 } n^{\varepsilon} \\ af(n/b) \le cf(n) \end{cases}$

找常數 $c < 1$

# Using the master method$_{2/3}$

- Example 3: $T(n) = 3T(n/4) + n \lg n$
  - For this recurrence, we have $a = 3$, $b = 4$, $f(n) = n \lg n$.
  - Thus, $n^{\log_b a} = n^{\log_4 3} = O(n^{0.793})$.
  - For sufficiently large $n$, $af(n/b) = 3(n/4) \lg(n/4) \le (3/4) n \lg n = cf(n)$ for $c = 3/4$.
  - Since $f(n) = \Omega(n^{\log_4 3 + \varepsilon})$ with $\varepsilon \approx 0.2$ and the regularity condition holds for $f(n)$ case 3 applies.
  - The solution is $T(n) = \Theta(n \lg n)$.

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- Strassen's algorithm for matrix multiplication

最大子陣列

# The maximum-subarray problem

- **Input:** An array $A[1\ldots n]$ of numbers.
    - Assume that some of the numbers are negative, because this problem is trivial when all numbers are nonnegative.
- **Output:** Indices $i$ and $j$ such that $A[i\ldots j]$ has the greatest sum of any contiguous subarray of array $A$.

找出i,j使得從i加到j的和是最大

- For example:
    - The subarray $A[8\ldots11]$, with sum 43, has the greatest sum of any contiguous subarray of array $A$. 回傳8和11

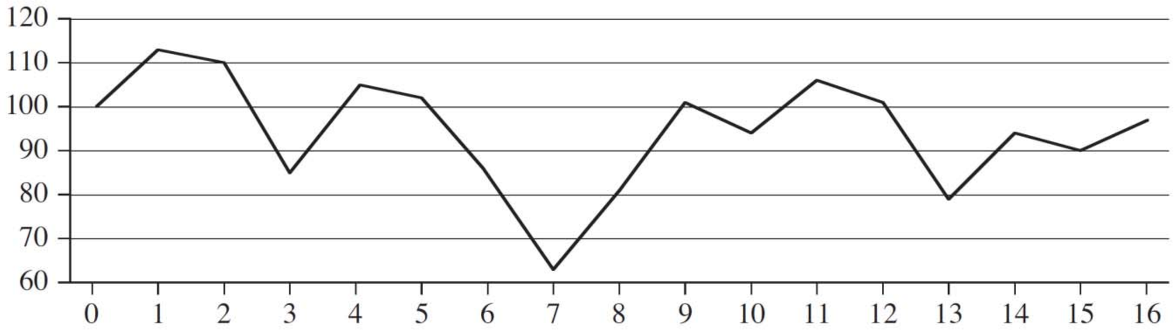
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $A$ | 13 | –3 | –25 | 20 | –3 | –16 | –23 | 18 | 20 | –7 | 12 | –5 | –22 | 15 | –4 | 7 |

maximum subarray

# An application

- You have the prices that a stock traded at over a period of *n* consecutive days.
- When should you have bought the stock? When should you have sold the stock? 第7天買，第11天賣賺最多

回傳8和11




| Day | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Price | 100 | 113 | 110 | 85 | 105 | 102 | 86 | 63 | 81 | 101 | 94 | 106 | 101 | 79 | 94 | 90 | 97 |
| Change | | 13 | −3 | −25 | 20 | −3 | −16 | −23 | 18 | 20 | −7 | 12 | −5 | −22 | 15 | −4 | 7 |

# Solving by brute-force algorithm

- **Brute-force algorithm:**
  - Check all $\binom{n}{2} + n = \theta(n^2)$ subarrays. 有 $\binom{n}{2}+n$ 種可能性
  - Organize the computation so that each subarray *A*[*i*...*j*] takes *O*(1) time.
  - So that the brute-force solution takes $\theta(n^2)$ time.

*A*[9...13] = -1

| | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|
| *A* | ... | -6 | 5 | -4 | 3 | 1 | -2 | 3 | -6 | ... |

*O*(1) time

算A[9...13]後算A[9...14]

只需O(1)時間

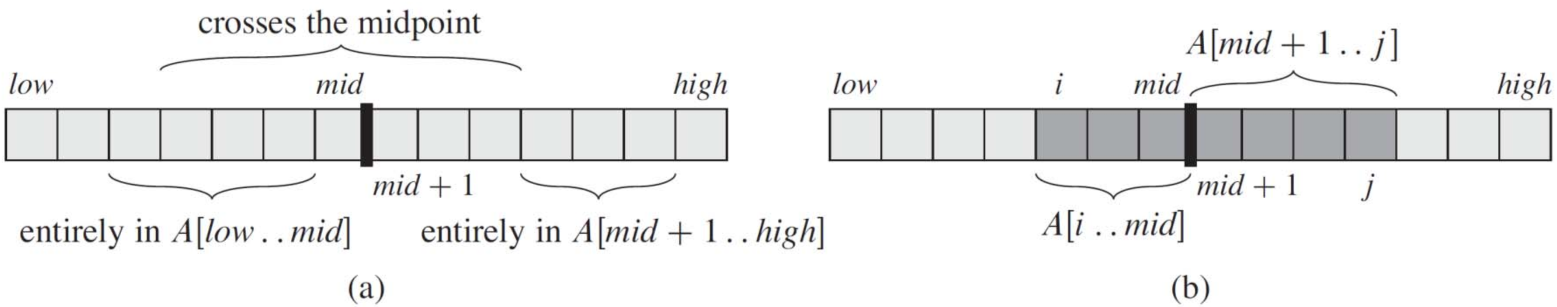
暴力法$\theta(n^2)$時間

*A*[9...14] = -3

| | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|
| *A* | ... | -6 | 5 | -4 | 3 | 1 | -2 | 3 | -6 | ... |

# Solving by divide-and-conquer

- **Divide** by splitting into two subarrays *A*[*low*...*mid*] and *A*[*mid*+1...*high* ], *mid* is the midpoint of *A*[*low*...*high* ].
  找出中間那一個
- **Conquer** by recursively finding a maximum subarrays of the two subarrays *A*[*low*...*mid*] and *A*[*mid*+1...*high* ].
  分别算左边最大和右边最大(子問題)
- **Combine** by finding a maximum subarray that crosses the midpoint, and using the best solution out of the three.
  找出跨越中間的最大，從三種可能性中取最佳(非子問題)

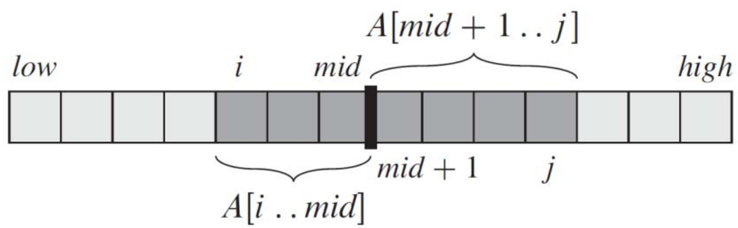


算跨中最大

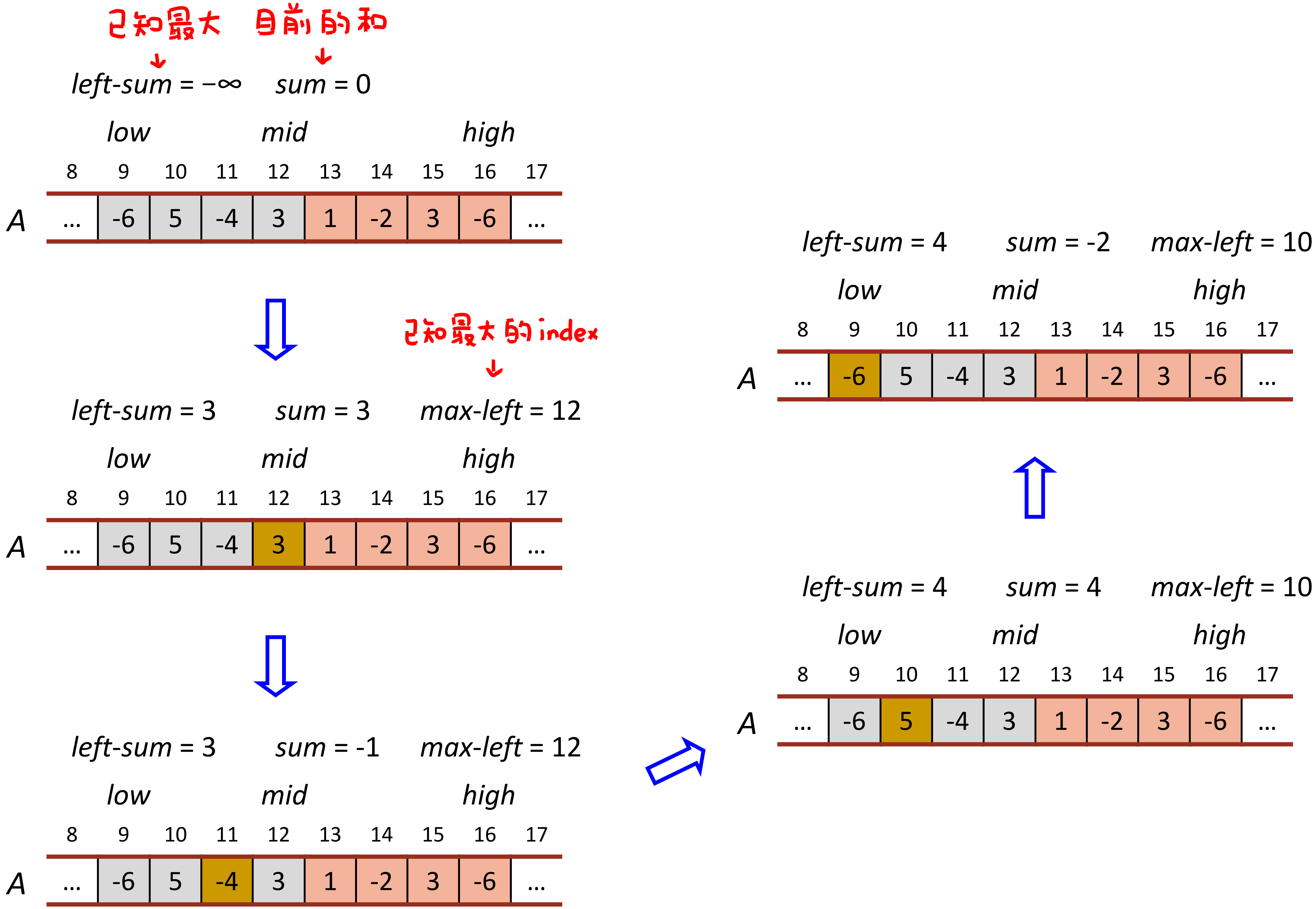
# Maximum subarray that crosses the midpoint

- **Not** a smaller instance of the original problem: has the added restriction that the subarray must cross the midpoint.
  不是原問題的子問題，因增加要跨中的條件
- Any subarray crossing the midpoint *A*[*mid*] is made of two subarrays *A*[*i*...*mid*] and *A*[*mid*+1...*j*].
  分別找出 A[i...mid] 的最大 和 A[mid+1...j] 的最大
- Find maximum subarrays of the form *A*[*i*...*mid*] and *A*[*mid*+1...*j*] and then combine them.

算跨中最大：算左半後綴+右半前綴




⇒將 i, j 找出



已知最大
目前的和
left-sum = −∞
sum = 0
low
mid
high
8 9 10 11 12 13 14 15 16 17
A
... -6 5 -4 3 1 -2 3 -6 ...
已知最大的index
left-sum = 3
sum = 3
max-left = 12
low
mid
high
8 9 10 11 12 13 14 15 16 17
A
... -6 5 -4 3 1 -2 3 -6 ...
left-sum = 3
sum = -1
max-left = 12
low
mid
high
8 9 10 11 12 13 14 15 16 17
A
... -6 5 -4 3 1 -2 3 -6 ...
left-sum = 4
sum = -2
max-left = 10
low
mid
high
8 9 10 11 12 13 14 15 16 17
A
... -6 5 -4 3 1 -2 3 -6 ...
left-sum = 4
sum = 4
max-left = 10
low
mid
high
8 9 10 11 12 13 14 15 16 17
A
... -6 5 -4 3 1 -2 3 -6 ...

# Find-Max-Crossing-Subarray

FIND-MAX-CROSSING-SUBARRAY(*A*, *low*, *mid*, *high*)

算左边

```
left-sum ← −∞
sum ← 0
for i ← mid downto low
        sum ← sum + A[i]
        if sum > left-sum
                left-sum ← sum
                max-left ← i
```

算右边

```
right-sum ← −∞
sum ← 0
for j ← mid +1 to high
        sum ← sum + A[j]
        if sum > right-sum
                right-sum ← sum
                max-right ← j
return (max-left, max-right, left-sum + right-sum)
```

$\Theta(1)$ 初始化

$\Theta(n)$ 從mid往左,一次加一個
如果目前的和>已知最大
→更新資訊

$\Theta(1)$ 初始化

$\Theta(n)$ 從mid往右,一次加一個
如果目前的和>已知最大
→更新資訊

**Time: $\Theta(n)$.**

# Divide-and-conquer procedure

FIND-MAXIMUM-SUBARRAY(*A*, *low*, *high*)

只有一個元素

**if** *high* == *low*

**return** (*low*, *high*, *A*[*low*]) // base case: only one element $\Theta(1)$

**else** *mid* ← $\lfloor (low + high)/2 \rfloor$

解決子問題

(*left-low*, *left-high*, *left-sum*) ← FIND-MAXIMUM-SUBARRAY(*A*, *low*, *mid*) $T(n/2)$ 算左邊最大

(*right-low*, *right-high*, *right-sum*) ← FIND-MAXIMUM-SUBARRAY(*A*, *mid*+1, *high*) $T(n/2)$ 算右邊最大

合併

(*cross-low*, *cross-high*, *cross-sum*) ← FIND-MAX-CROSSING-SUBARRAY(*A*, *low*, *mid*, *high*) $\Theta(n)$ 算跨中最大

**if** *left-sum* ≥ *right-sum* and *left-sum* ≥ *cross-sum*

**return** (*left-low*, *left-high*, *left-sum*)

**elseif** *right-sum* ≥ *left-sum* and *right-sum* ≥ *cross-sum*

**return** (*right-low*, *right-high*, *right-sum*)

**else return** (*cross-low*, *cross-high*, *cross-sum*)

$\Theta(1)$

回傳3種可能性中最大的

**Initial call :** FIND-MAXIMUM-SUBARRAY(*A*, 1, *n*)

# Analyzing maximum-subarray

- For simplicity, assume that $n$ is a power of 2.

$$T(n) = \begin{cases} \theta(1) & \text{if } n = 1, \\ 2T(n/2) + \theta(n) & \text{otherwise.} \end{cases}$$

$f(n) =$ divide + combine
$\theta(1)$ $\theta(n)$

  - The base case occurs when $n = 1$.
  - **Divide**: compute the middle of the subarray, $D(n) = \Theta(1)$.
  - **Conquer**: Recursively solve 2 subproblems, each of size $n/2$.
  - **Combine**: Combining consists of calling FIND-MAX-CROSSING-SUBARRAY, which takes $\Theta(n)$ time, and a constant number of constant-time tests $\Rightarrow C(n) = \Theta(n) + \Theta(1)$ time for combining.

- By using master method, we have $T(n) = \Theta(n \lg n)$.

大師方法 case 2

# Outline

- The substitution method
- The recursion-tree method
- The master method
- The maximum-subarray problem
- **Strassen's algorithm for matrix multiplication**

# Matrix multiplication

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}$$

- **Input:** Two $n \times n$ matrices, $A = (a_{ij})$ and $B = (b_{ij})$.
- **Output:** $n \times n$ matrix, $C = (c_{ij})$, where $C = A \cdot B$,i.e.,

$$c_{ji} = \sum_{k=1}^{n} a_{ik} b_{kj} \qquad \text{for } i, j = 1, 2, \dots, n.$$

  - Need to compute $n^2$ entries of $C$.
  - Each entry is the sum of $n$ values.

SQUARE-MATRIX-MULTIPLY ($A$, $B$)

$n \leftarrow A.rows$<br>
*let C be a new* $n \times n$ *matrix*<br>
**for** $i \leftarrow 1$ **to** $n$<br>
    **for** $j \leftarrow 1$ **to** $n$<br>
        $c_{ij} \leftarrow 0$<br>
        **for** $k \leftarrow 1$ **to** $n$<br>
            $c_{ij} \leftarrow c_{ij} + a_{ik} \cdot b_{kj}$<br>
**Return** $C$

$c_{11} = a_{11} \times b_{11} + a_{12} \times b_{21} + \dots + a_{1n} \times b_{n1}$

C有$n^2$個元素，每一個皆為n個值相加

**Time: $\Theta(n^3)$.**

# Simple divide-and-conquer method

- Partition each of $A, B$ and $C$ into four $n/2 \times n/2$ matrices, so that we rewrite the equation $C = A \cdot B$ as C=A·B的另一種看法

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \cdot \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

- The four corresponding equations are:
  - $C_{11} = A_{11} \cdot B_{11} + A_{12} \cdot B_{21}$,
  - $C_{12} = A_{11} \cdot B_{12} + A_{12} \cdot B_{22}$,
  - $C_{21} = A_{21} \cdot B_{11} + A_{22} \cdot B_{21}$,
  - $C_{22} = A_{21} \cdot B_{12} + A_{22} \cdot B_{22}$.

8個子問題，每一個大小是 $\frac{n}{2}$

$\boxed{A_{11} \cdot B_{11}} + \boxed{A_{12} \cdot B_{21}}$

相加需要 $\Theta(n^2)$ 的時間

- Each of these equations multiplies two $n/2 \times n/2$ matrices and then adds their $n/2 \times n/2$ products.

# Procedure of matrix-multiply-recursive

SQUARE-MATRIX-MULTIPLY-RECURSIVE ($A$, $B$)

```
n ← A.rows
let C be a new n × n matrix
if n == 1                          // base case: only one element
    c11 ← a11 · b11
else partition each of A, B and C into four n/2 × n/2 matrices
    C11 ← SQUARE-MATRIX-MULTIPLY-RECURSIVE (A11, B11)
          + SQUARE-MATRIX-MULTIPLY-RECURSIVE (A12, B21)
    C12 ← SQUARE-MATRIX-MULTIPLY-RECURSIVE (A11, B12)
          + SQUARE-MATRIX-MULTIPLY-RECURSIVE (A12, B22)
    C21 ← SQUARE-MATRIX-MULTIPLY-RECURSIVE (A21, B11)
          + SQUARE-MATRIX-MULTIPLY-RECURSIVE (A22, B21)
    C22 ← SQUARE-MATRIX-MULTIPLY-RECURSIVE (A21, B12)
          + SQUARE-MATRIX-MULTIPLY-RECURSIVE (A22, B22)
return C
```

Lines 1–2: $\Theta(1)$

Lines 3–4: $\Theta(1)$

Lines 5–9: $8T(n/2) + \Theta(n^2)$

$f(n) = \underbrace{\text{divide}}_{\Theta(1)} + \underbrace{\text{combine}}_{\Theta(n^2)}$

8個子問題，每一個大小是$\frac{n}{2}$

$\boxed{A_{11} \cdot B_{11}} + \boxed{A_{12} \cdot B_{21}}$

相加需要$\Theta(n^2)$的時間

# Analyzing

- For simplicity, assume that $n$ is a power of 2.

$$T(n)=\begin{cases}\theta(1) & \text{if } n=1,\\ 8T(n/2)+\theta(n^2) & \text{otherwise.}\end{cases}$$

$f(n) = \text{divide} + \text{combine}$
$\theta(1) \quad \theta(n^2)$

- The base case occurs when $n$ = 1.
- **Divide**: Partition $A, B$ and $C$ into four $n/2 \times n/2$ matrices by index calculation takes Θ(1), $D(n)$ = Θ(1). 8個子問題，每一個大小是$\frac{n}{2}$
- **Conquer**: Recursively solve 8 subproblems, each of size $n/2$.
- **Combine**: Combining takes $\Theta(n^2)$ time to add $n/2 \times n/2$ matrices four times. ⇒ $C(n) = \Theta(n^2)$ time for combining. $\boxed{A_{11} \cdot B_{11}} + \boxed{A_{12} \cdot B_{21}}$ 相加需要$\theta(n^2)$的時間
- By using master method, we have $T(n) = \Theta(n^3)$.

大師方法 case 1

strassen's algorithm 核心思想
把原先矩陣各4等分，再把原先8個子矩陣乘法用加法整併其中2項
→減少乘法運算

# Strassen's method

分割
- Step 1: partition each of $A, B$ and $C$ into four $n/2 \times n/2$ matrices. **Time: Θ(1).** 產生 $A_{11}, A_{12}, A_{21}, A_{22}$ $B_{11}, B_{12}, B_{21}, B_{22}$ $C_{11}, C_{12}, C_{21}, C_{22}$ Divide 產生子矩陣
- Step 2: create 10 matrices $S_1, S_2, \ldots, S_{10}$, each of which is $n/2 \times n/2$ and is the sum or difference of two matrices created in step 1. **Time: Θ($n^2$).** 產生 $S_1, S_2 \ldots S_{10}$ 這10個矩陣
  P.47 如何產生矩陣S

擊破
- Step 3: using the submatrices created in Step 1 and the 10 matrices created in step 2, recursively compute seven matrix products $P_1, P_2, \ldots, P_7$. Each matrix $P_i$ is $n/2 \times n/2$. **Time: 7$T(n/2)$.**
  解決 $P_1, P_2 \ldots P_7$，這7個子問題 P.48 解決子問題P

合併
- Step 4: Compute the desired submatrices $C_{11}, C_{12}, C_{21}, C_{22}$ of the result matrix $C$ by adding and subtracting various combinations of the $P_i$ matrices. **Time: Θ($n^2$).** 由 $P_1, P_2 \ldots P_7$ 的解產生矩陣 C

# Step 2: create the 10 matrices

- $S_1 = B_{12} - B_{22}$ ,
- $S_2 = A_{11} + A_{12}$ ,
- $S_3 = A_{21} + A_{22}$ ,
- $S_4 = B_{21} - B_{11}$ ,
- $S_5 = A_{11} + A_{22}$ ,
- $S_6 = B_{11} + B_{22}$ ,
- $S_7 = A_{12} - A_{22}$ ,
- $S_8 = B_{21} + B_{22}$ ,
- $S_9 = A_{11} - A_{21}$ ,
- $S_{10} = B_{11} + B_{12}$ . **Time: $\Theta(n^2)$.**

# Step 3: create the 7 matrices

- $P_1 = A_{11} \cdot S_1 = A_{11} \cdot B_{12} - A_{11} \cdot B_{22}$ ,
- $P_2 = S_2 \cdot B_{22} = A_{11} \cdot B_{22} + A_{12} \cdot B_{22}$ ,
- $P_3 = S_3 \cdot B_{11} = A_{21} \cdot B_{11} + A_{22} \cdot B_{11}$ ,
- $P_4 = A_{22} \cdot S_4 = A_{22} \cdot B_{21} - A_{22} \cdot B_{11}$ ,
- $P_5 = S_5 \cdot S_6 = A_{11} \cdot B_{11} + A_{11} \cdot B_{22} + A_{22} \cdot B_{11} + A_{22} \cdot B_{22}$,
- $P_6 = S_7 \cdot S_8 = A_{12} \cdot B_{21} + A_{12} \cdot B_{22} - A_{22} \cdot B_{21} - A_{22} \cdot B_{22}$,
- $P_7 = S_9 \cdot S_{10} = A_{11} \cdot B_{11} + A_{11} \cdot B_{12} - A_{21} \cdot B_{11} - A_{21} \cdot B_{12}$.

**Time: 7*T*(*n*/2).**

## Step 4: construct submatrices of $C$

- $C_{11} = P_5 + P_4 - P_2 + P_6$ ,
- $C_{12} = P_1 + P_2$ ,
- $C_{21} = P_3 + P_4$ ,
- $C_{22} = P_5 + P_1 - P_3 - P_7$ .

**Time: $\Theta(n^2)$.**

# Analyzing

- For simplicity, assume that $n$ is a power of 2.

$$T(n) = \begin{cases} \theta(1) & \text{if } n = 1, \\ 7T(n/2) + \theta(n^2) & \text{otherwise.} \end{cases}$$

$f(n) = \underbrace{\text{divide}}_{\theta(n^2)} + \underbrace{\text{combine}}_{\theta(n^2)}$

  - The base case occurs when $n$ = 1.
  - **Divide**: Partition $A, B$ and $C$ into four $n/2 \times n/2$ matrices by index calculation takes $\Theta(1)$. Creating the matrices $S_1, S_2, \ldots, S_{10}$, each of which is $n/2 \times n/2$ takes $\Theta(n^2)$, $D(n) = \Theta(1) + \Theta(n^2) = \Theta(n^2)$ .
  7個子問題，每一個大小是$\frac{n}{2}$
  - **Conquer**: Recursively solve 7 subproblems, each of size $n/2$.
  - **Combine**: Combining takes $\Theta(n^2)$ time to add and subtract $n/2 \times n/2$ matrices. $\Rightarrow C(n) = \Theta(n^2)$ time for combining.
- By using master method, we have $T(n) = \Theta(n^{\lg 7})$.
  大師方法 case 1